

Precalc Lesson 1.4 Homework

Answer Key

1. Domain: $[0, \infty)$
Range: $[0, \infty)$
2. a. $y = x^2$
b. $y = |x|$
3. $y = x^2$ is increasing on $[0, \infty)$ and decreasing on $(-\infty, 0]$. There are no intervals on which $y = x^2$ is constant.
4. $4f(-2) + g(-2) = -8.5$
5. A
6. a. Greatest output: $y = \frac{1}{x}$
b. Least output: $y = x^3$
7. $y = \sqrt{x}$
8. Many answers are possible as long as $k < 0$ or $0 < k < 1$.
9. Always.
When $x < 0$, $|x| > 0$ and when $x \geq 0$, $|x| = x$.
10. The parent function $y = x$ has a slope of 1 and a y -intercept of 0, whereas g has a slope of a and a y -intercept of b . The domain and range of g are the same as the parent function, all real numbers.